

Piotr Koper

Senior Data Engineer · Platform & Infrastructure

West Midlands, UK · piotr.koper.v0@gmail.com · p-y-t-h-e-c.github.io/portfolio

PROFESSIONAL SUMMARY

Senior Data Engineer with 4+ years at Jaguar Land Rover's Data & AI Office, specialising in data platform infrastructure, CI/CD pipeline engineering, and cloud infrastructure on GCP. Progressed from pipeline development and data products into technical leadership — currently leading working groups across GitLab CI/CD, Terraform, and Docker image management. Background in logistics and operations management across major UK brands brings strong domain knowledge and stakeholder communication skills developed at scale. Open to Data Engineering, Platform Engineering, DevOps, and Analytics Engineering roles where infrastructure depth and engineering rigour are valued.

TECHNICAL SKILLS

Programming Languages: Python · SQL · Bash · HCL · YAML

Cloud Platforms & Services: GCP — BigQuery · GCS · Cloud Run · Cloud Functions · Cloud SQL · Artifact Registry · AWS · Snowflake

Infrastructure & IaC: Terraform · Docker · Kaniko · Podman · Kubernetes · Helm

CI/CD & DevOps: GitLab CI/CD · GitHub Actions · GitLab Components · Pipeline Templating · Container Scanning

Data Orchestration & Transformation: Dagster · Apache Airflow (Cloud Composer / Astronomer) · dbt · Dataform

Pipeline Patterns: ETL / ELT · Pipeline Architecture · Data Modelling · Data Contracts · APIs

Security & Identity: Workload Identity Federation · IAM · Secret Management

Developer Tooling: Git · Linux · Pre-commit · uv · Grafana

PROFESSIONAL EXPERIENCE

Senior Data Engineer · Jaguar Land Rover · 2022 — Present

Part of the core Data Engineering team within JLR's Data & AI Office (DAIO), focused on infrastructure, developer tooling, and engineering standards. Progressed from pipeline development and data products into platform engineering, taking on technical leadership across multiple working groups.

Data Pipelines & Products

- Designed, developed and delivered ETL and ELT data pipelines for high-volume data applications using Python and SQL, incorporating automated data testing to ensure continuous data quality in deployment
- Evaluated Data Contract standards through hands-on testing, including dbt integration — and made a merged contribution to the Open Data Contract Standard (ODCS) open source project, shipped in v3.1.0

CI/CD Working Group — Lead

Own and maintain semantically versioned GitLab CI/CD pipeline templates across the Data & AI Office and wider organisation, acting as one of the primary merge request reviewers and approvers.

- Develop new templates and drive reusability, code quality, security, and consistency — enabling multiple teams across the enterprise to validate and deploy code quickly with enforced standards
- Refactored multiple existing pipelines into a modular structure of reusable components, default templates, and pipeline definitions — covering Docker image builds, container scanning, Cloud Run, Cloud Functions, and other deployment targets
- Led the Workload Identity Federation rollout, designing an authentication template that replaces long-lived credentials in CI/CD variables with short-lived OIDC tokens while remaining backward compatible with legacy Service Account Key projects — materially improving security posture across a high volume of daily pipeline runs
- Designed and delivered bespoke CI/CD pipelines for the Dagster and Astronomer Airflow orchestration platforms, enabling SaaS integration with automated, controlled deployments

Terraform Working Group — Lead

Lead the Terraform working group, owning module standards, provisioning patterns, and infrastructure-as-code practices across the data platform.

- Build and maintain reusable Terraform modules that let teams provision cloud resources consistently via CI/CD, without bespoke infrastructure work per project
- Provisioned the Workload Identity Federation infrastructure — Pool, Provider, and bindings — underpinning keyless GitLab to GCP authentication across the organisation
- Refactored the Notifications System Terraform module end to end — Cloud Run services behind API Gateway, private Cloud SQL PostgreSQL with enforced SSL and automated backups, Memorystore Redis, and Secret Manager with least-privilege IAM access, all over private VPC networking

Docker Images Working Group — Lead

Lead the working group responsible for maintaining and improving standardised Docker images stored in GCP Artifact Registry, used by the DAIO and other teams.

- Maintain and improve custom runtime images for GitLab runners with the required tooling and dependencies pre-installed — removing runtime install steps from CI jobs, simplifying pipeline definitions, and reducing pipeline execution time
- Keep images current and secure through regular base image version updates and package maintenance

Standards, Documentation & Best Practices

Develop and maintain engineering standards, guides, and best practices adopted by technical and non-technical teams across the organisation — written as Docs as Code, versioned and reviewed alongside the code they describe.

- Authored end-to-end documentation for key platform initiatives — including the Workload Identity Federation architecture, implementation and migration guide, the Notifications System, Terraform modules, test projects, Dagster and Astronomer Airflow guides, and CI/CD components and pipeline templates
- Primary developer of standardised project setup scripts in Bash (Linux) and PowerShell (Windows), paired with pre-commit hooks for linting, formatting, and validation — promoting consistent development practices across teams

Notable Projects

- **Python Task Orchestrator Evaluation (Lead)** — structured evaluation of two enterprise orchestration platforms, building end-to-end data pipelines covering data validation, CI/CD integration, event-triggered execution, cross-platform authentication, and integrations with GCP, dbt, Dataform, and Snowflake. The formal evaluation report was adopted as the technical recommendation to the Enterprise Technology Board
- **Notifications System** — core developer of a GCP-based notifications system routing messages to email, Slack, and Jira based on subscription behaviour and API calls during pipeline execution
- **GitLab Instance Migration** — part of the team that designed and provisioned the Kubernetes cluster, namespace, and Helm module for self-hosted GitLab runners during a full instance migration, managed entirely via Terraform
- **Enterprise Data Platform Investigation** — contributed to a cross-functional investigation with the Enterprise Architect into the target layout and usage patterns for JLR's Enterprise Data Platform

CPAT Lead & Planner · [Jaguar Land Rover](#) · 2018 — 2022

Cross-functional programme role managing the Capacity, Parts Availability and Timing assessment process across JLR vehicle programmes.

- Built a redesigned CPAT Input Matrix addressing existing error states, improving data input control and reducing submission errors across programme teams
- Developed an automated WIP file creation process saving approximately 12 hours per file, and an eCAR data scrape tool that replaced a manual weekly data extraction

Planning Manager & Logistics Operations · [Halfords](#) · [Aston Martin Lagonda \(DHL\)](#) · [DHL](#) · 2007 — 2018

Over a decade of progressive operations management across major UK distribution and manufacturing environments — Halfords (1.4M units/week), Aston Martin Lagonda, and DHL.

- Consistently built tooling and automation to replace manual processes — a Logistics Performance Tool tracking volumes, hours and costs against budget, automated planning tools, and redesigned operational workflows
- Managed teams of up to 130 people; selected for the competitive DHL Head Start management programme — one of 28 accepted from hundreds of UK and Ireland applicants

PERSONAL PROJECTS

Two personal data engineering projects on a shared infrastructure pattern: Dagster pipelines, PostgreSQL for run state, Docker Compose orchestration, and automated deployment to an Oracle Cloud VM via GitHub Actions — build, push to a container registry, SSH deploy, and restart with zero manual steps.

Solis Battery Manager · github.com/p-y-t-h-e-c/solis_energy_manager · 2025

- Scheduled Dagster pipeline ingesting solar inverter data from the Solis Cloud API (HMAC-signed auth), validating it, and emailing daily battery charge, PV generation, and export summaries via Pingram
- Multi-platform Docker images (amd64/arm64) published to GitHub Container Registry on every push to main; backed by full project documentation

Octopus Energy Dashboard · [Personal Project](#) · 2023

- Dagster pipeline ingesting variable tariff and consumption data from the Octopus Energy API into PostgreSQL, surfaced through a Grafana dashboard with day-on-day and year-over-year cost comparisons

EDUCATION, CERTIFICATIONS & LANGUAGES

MSc Management and Production · [Technical University of Łódź, Poland](#) · 2001 — 2007

Majoring in Systems and Information Technologies

MSc Information and Communication Technology for Engineers · [Coventry University, UK](#) · 2005 — 2006

Analysis and Design of Engineering Information Systems · Project Management · Software Design for Engineers

GCP Professional Data Engineer · Google Cloud · 2021 (expired — experience remains current)

Polish — native · English — fluent